

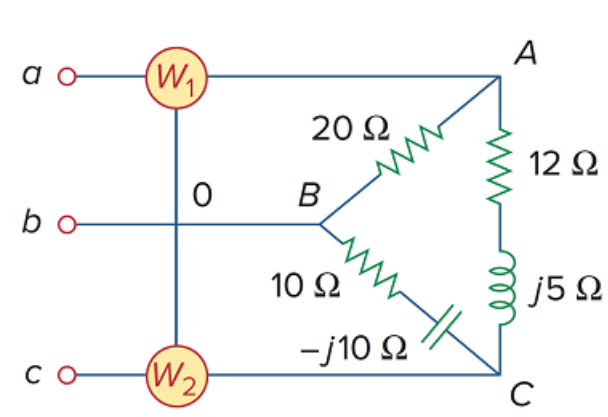
Problem

In Fig. 12.73, two wattmeters are properly connected to the unbalanced load supplied by a balanced source such that $\mathbf{V}_{ab} = 208 \angle 0^\circ \text{ V}$ with positive phase sequence.

(a) Determine the reading of each wattmeter.

(b) Calculate the total apparent power absorbed by the load.

Figure 12.73



Step-by-step solution

Step 1 of 3

(A) If $V_{ab} = 208 \angle 0^\circ$, $V_{bc} = 208 \angle -120^\circ$, $V_{ca} = 208 \angle 120^\circ$

$$I_{AB} = \frac{V_{ab}}{Z_{AB}} = \frac{208 \angle 0^\circ}{20} = 10.4 \angle 0^\circ$$

$$I_{BC} = \frac{V_{bc}}{Z_{BC}} = \frac{208 \angle -120^\circ}{10\sqrt{2} \angle -45^\circ} = 14.708 \angle -75^\circ$$

$$I_{CA} = \frac{V_{ca}}{Z_{CA}} = \frac{208 \angle 120^\circ}{13 \angle 22.62^\circ} = 16 \angle 97.38^\circ$$

$$I_{aA} = I_{AB} - I_{CA} = 10.4 \angle 0^\circ - 16 \angle 97.38^\circ$$

$$I_{aA} = 10.4 + 2.058 - j15.867$$

$$I_{aA} = 20.171 \angle -51.87^\circ$$

$$I_{cC} = I_{CA} - I_{BC} = 16 \angle 97.83^\circ - 14.708 \angle -75^\circ$$

$$I_{cC} = 30.64 \angle 101.03^\circ$$

$$P_1 = |V_{ab}| |I_{aA}| \cos(\theta_{V_{ab}} - \theta_{I_{aA}})$$

$$P_1 = (208)(20.175) \cos(0^\circ - 51.87^\circ) = 2590 \text{ W}$$

$$P_2 = |V_{cb}| |I_{cC}| \cos(\theta_{V_{cb}} - \theta_{I_{cC}})$$

Step 2 of 3

But,

$$V_{cb} = -V_{bc} = 208 \angle 60^\circ$$

$$P_2 = (208) (30.64) \cos(60^\circ - 101.03^\circ) = 4808 \text{ W} .$$

Step 3 of 3

$$(B) P_T = P_1 + P_2 = 7398.17 \text{ W}$$

$$Q_T = \sqrt{3}(P_2 - P_1) = 3840.25 \text{ VAR}$$

$$S_T = P_T + jQ_T = 7398.17 + j3840.25 \text{ VA}$$

$$S_T = |S_T| = 8335 \text{ VA} .$$